

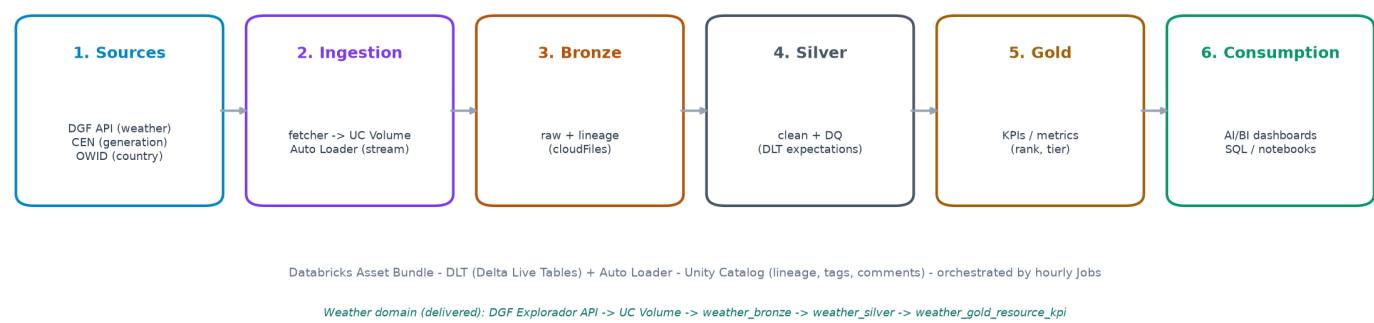
Renewable-Energy PoC - Technical Documentation

Medallion lakehouse on Databricks | 2026-07-02

1. Architecture

Databricks Asset Bundle with DLT pipelines on serverless, Auto Loader (cloudFiles) streaming ingestion from Unity Catalog Volumes, and Unity Catalog governance. Flow: Sources -> Ingestion -> Bronze -> Silver -> Gold -> Consumption, orchestrated by hourly Jobs.

Renewable-Energy PoC - Data Architecture (Medallion on Databricks)



2. Domains & data sources

- Resource (renewable_energy_chile): generation per plant (coordinado/real/reducciones). Source: CEN / Coordinador Electrico Nacional.
- Conglomerate (renewable_conglomerate_energy): country stats (Entity/Year). Source: Our World in Data.
- Weather (delivered): solar irradiance + wind. Source: DGF Explorador. The DGF owns the resource/weather data, not the generation.

3. Weather pipeline (this deliverable)

- Acquisition: scripts/dgf_explorador_fetcher.py calls the open python-router endpoint (no auth), lands JSON in the UC Volume; retries transient empty responses.
- Bronze: weather_bronze - Auto Loader streaming table (cloudFiles json); raw + file lineage.
- Silver: weather_silver - flatten/type per plant; keep latest per plant; DLT expectations (known resource, Chile-bounds coords, has a metric).
- Gold: weather_gold_resource_kpi - per-plant KPI: metric (solar GHI / wind m/s), rank, A/B/C tier.
- Orchestration: weather_streaming_job (hourly) = fetch_weather (serverless python) -> weather_medallion (DLT).

4. Data quality, lineage & glossary (Databricks-native)

- Data quality / contracts: DLT expectations (event log) + Unity Catalog constraints.
- Lineage: Unity Catalog automatic table + column lineage.
- Glossary: UC tags + column comments (100% documented) + docs/glossary.md.

5. Deploy & run

databricks bundle deploy -t dev --var=dev_catalog=workspace

databricks bundle run weather_streaming_job -t dev --var=dev_catalog=workspace

Prod deploys are managed by the project lead (prod target scoped to their workspace).

Explore / test the results

- Job (fetch -> medallion):

<https://dbc-54b27bae-2e91.cloud.databricks.com/jobs/503330163541320?o=7474645896934260>

- **Pipeline DLT (bronze->silver->gold):**

<https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/c108b287-98cc-43d6-86d6-5b63a9e6b4ae?o=7474645896934260>

- **Gold table (result):**

https://dbc-54b27bae-2e91.cloud.databricks.com/explore/data/workspace/dev_fuad_onate_renewable_gold_energy_chile/weather_gold_resource_kpi?o=7474645896934260

- **Code (PR #24):**

https://github.com/oxiboy/poc_databricks_evaluateserve/pull/24